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UNIT 2 / 第二单元

Chain Polymerization / 链式聚合

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Paragraph 1 / 第一段

English

Many olefinic and vinyl unsaturated compounds are able to form chain-like macromolecules through elimination of the double bond, a phenomenon first recognized by Staudinger. Diolefins polymerize in the same manner, however, only one of the two double bonds is eliminated. Such reactions occur through the initial addition of a monomer molecule to an initiator radical or an initiator ion, by which the active state is transferred from the initiator to the added monomer. ① In the same way, by means of a chain reaction, one monomer molecule after the other is added (2000~20000 monomers per second) until the active state is terminated through a different type of reaction. The polymerization is a chain reaction in two ways: because of the reaction kinetics and because as a reaction product one

obtains a chain molecule. The length of the chain molecule is proportional to the kinetic chain length.

中文翻译

许多烯烃类和乙烯基不饱和化合物都能够通过双键的消除形成链状大分子，这一现象最早由 Staudinger 认识到。二烯烃也以同样的方式发生聚合，不过两个双键中通常只有一个被消除。此类反应首先通过单体分子与引发剂自由基或引发剂离子的加成而发生，由此活性状态从引发剂转移到已加成的单体上。① 同样地，通过链式反应，一个单体分子接着一个单体分子连续加成（每秒可加成 2000～20000 个单体分子），直到活性状态通过另一种类型的反应而被终止。聚合在两种意义上都属于链反应：一是从反应动力学上看如此，二是因为所得反应产物本身就是链状分子。链分子的长度与动力学链长成正比。

Paragraph 2 / 第二段

English

One can summarize the process as follows (R • is equal to the initiator radical):

中文翻译

这一过程可以概括如下（R• 表示引发剂自由基）：

Reaction scheme / 反应示意

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R• + CH2=CHCl + CH2=CHCl + CH2=CHCl + ···  →  R–CH2–CH(Cl)–CH2–CH(Cl)–CH2–CH(Cl)–
···
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中文说明

以上示意图以氯乙烯为例，表示链引发后，单体分子不断加成到活性链端上，逐步形成聚氯乙烯链。

Paragraph 3 / 第三段

English

One thus obtains polyvinylchloride from vinylchloride, or polystyrene from styrene, or polyethylene from ethylene, etc.

中文翻译

因此，可以由氯乙烯得到聚氯乙烯，由苯乙烯得到聚苯乙烯，由乙烯得到聚乙烯，等等。

Paragraph 4 / 第四段

English

The length of the chain molecules, measured by means of the degree of polymerization, can be varied over a large range through selection of suitable reaction conditions. Usually, with commercially prepared and utilized polymers, the degree of polymerization lies in the range of 1000 to 5000, but in many cases it can be below 500 and over 10000. This should not be interpreted to mean that all molecules of a certain polymeric material consist of 500, or 1000, or 5000 monomer units. In almost all cases, the polymeric material consists of a mixture of polymer molecules of different degrees of polymerization.

中文翻译

链分子的长度可用聚合度来衡量，并且能够通过选择适当的反应条件在很大范围内加以调节。通常，对于工业上制备和使用的聚合物，其聚合度处于 1000 到 5000 的范围内；但在许多情况下，也可能低于 500，或高于 10000。这并不意味着某一种高分子材料中的所有分子都恰好由 500、1000 或 5000 个单体单元组成。实际上，几乎在所有情况下，高分子材料都是由不同聚合度的聚合物分子所组成的混合物。

Paragraph 5 / 第五段

English

Polymerization, a chain reaction, occurs according to the same mechanism as the well-known chlorine-hydrogen reaction and the decomposition of phosgene.

中文翻译

聚合是一种链反应，其发生机理与人们熟知的氯—氢反应以及光气的分解反应相同。

Paragraph 6 / 第六段

English

The initiation reaction, which is the activation process of the double bond, can be brought about by heating, irradiation, ultrasonics, or initiators. The initiation of the chain reaction can be observed most clearly with radical or ionic initiators. ② These are energy-rich compounds which can add suitable unsaturated compounds (monomers) and maintain the activated radical or ionic state so that further monomer molecules can be added in the same manner. ③ For the individual steps of the growth reaction one needs only a relatively small activation energy and therefore through a single activation step (the actual initiation reaction) a large number of olefin molecules are converted, as is implied by the term "chain reaction". ④

中文翻译

引发反应，也就是双键的活化过程，可以通过加热、辐照、超声波或引发剂来实现。用自由基引发剂或离子引发剂时，链反应的引发过程表现得最为清楚。② 这些都是富含能量的化合物，能够与适当的不饱和化合物（单体）发生加成，并保持活化的自由基状态或离子状态，从而使后续单体分子也能以同样方式继续加成。③ 对于增长反应的各个单独步骤来说，只需要相对较小的活化能，因此仅通过一个活化步骤（即真正的引发反应），就可以使大量烯烃分子发生转化，这也正是“链反应”这一术语所蕴含的意思。④

Paragraph 7 / 第七段

English

Because very small amounts of the initiator bring about the formation of a large amount of polymeric material (1 : 1000 to 1: 10000), it is possible to regard polymerization from a superficial point of view as a catalytic reaction. For this reason, the initiators used in polymerization reactions are often designated as polymerization catalysts, even though, in the strictest sense, they are not true catalysts because the polymerization initiator enters into the reaction as a real partner and can be found chemically bound in the reaction product, i. e., the

polymer. In addition to the ionic and radical initiators there are now metal complex initiators (which can be obtained, for example, by the reaction of titanium tetrachloride or titanium trichloride with aluminum alkyls), which play an important role in polymerization reactions (Ziegler catalysts). The mechanism of their catalytic action is not yet completely clear.

中文翻译

由于极少量的引发剂就能够促成大量高分子材料的形成（比例约为 1:1000 到 1:10000），因此从表面上看，可以把聚合反应视为一种催化反应。正因为如此，聚合反应中所用的引发剂常常被称为聚合催化剂；然而，从最严格的意义上说，它们并不是真正的催化剂，因为聚合引发剂会作为真实的反应参与者进入反应，并且能够以化学键结合的形式存在于反应产物中，也就是存在于聚合物中。除离子型和自由基型引发剂之外，现在还有金属配合物引发剂（例如可由四氯化钛或三氯化钛与烷基铝反应得到），它们在聚合反应中起着重要作用（即齐格勒催化剂，Ziegler catalysts）。至于它们催化作用的机理，目前还没有被完全弄清。

Terms for Preview / 预习高频术语

- chain polymerization —— 链式聚合
- olefinic compounds —— 烯烃类化合物
- vinyl unsaturated compounds —— 乙烯基不饱和化合物
- diolefins —— 二烯烃
- initiator radical —— 引发剂自由基
- initiator ion —— 引发剂离子
- active state —— 活性状态
- degree of polymerization —— 聚合度
- kinetic chain length —— 动力学链长
- initiation reaction —— 引发反应
- growth reaction / propagation —— 增长反应 / 链增长
- radical initiator —— 自由基引发剂
- ionic initiator —— 离子引发剂
- polymerization catalysts —— 聚合催化剂
- metal complex initiators —— 金属配合物引发剂
- Ziegler catalysts —— 齐格勒催化剂

One-Sentence Summary / 一句话总结

English

Chain polymerization is a process in which unsaturated monomers successively add to an active center to form long-chain macromolecules.

中文翻译

链式聚合是指不饱和单体连续加成到活性中心上，从而形成长链大分子的过程。